

The Eaton–Kortum Model, Fully Derived

A step-by-step companion to “Preliminaries to Eaton–Kortum (2002)” and
“Eaton–Kortum (2002)”

Study notes

Contents

I	The Technology Block	3
1	An idea drawn from a Pareto distribution	3
1.1	Truncation from below preserves the Pareto shape	4
2	The accumulation of ideas	5
2.1	The Poisson limit	5
3	The most efficient idea: the Fréchet distribution	6
4	Putting the technology block into a Ricardian model	8
4.1	Setup: DFS with probability theory	8
4.2	Distribution of relative productivity	9
4.3	Selection probabilities	9
4.4	Distribution of prices	10
4.5	The price index	11
4.6	Equilibrium (two countries)	12
II	Eaton–Kortum (2002)	12
5	Background: why EK?	12
6	Setup with N countries	13
7	Prices with N countries	13
8	Trade shares with N countries	13
9	EK <i>vs.</i> Armington: what margin does trade adjust on?	14
10	Estimation	15
10.1	The estimating equation	15
10.2	Strategy 1: unpack S_i with technology proxies	16
10.3	Strategy 2: retail price data	16

11 Calibration and counterfactuals	17
A Answers to the checkpoint questions	17

Part I

The Technology Block

1 An idea drawn from a Pareto distribution

The whole EK machine starts from one modeling choice: think of *technology* not as a fixed number, but as the best of many random “ideas.” We first need a distribution for the quality of a single idea.

Definition: Pareto distribution

An idea for producing a good has **efficiency** Z : how much output one unit of input produces. Z is a random variable with CDF

$$F(z) = \Pr(Z \leq z) = 1 - \left(\frac{z}{z_0}\right)^{-\theta}, \quad z \geq z_0, \quad (1)$$

with **scale** $z_0 > 0$ (the worst possible idea) and **shape** $\theta > 0$ (how thin the upper tail is).

Intuition: Why Pareto?

The Pareto is the canonical “fat-tailed” distribution: the survival probability $\Pr(Z \geq z) = (z/z_0)^{-\theta}$ decays like a *power* of z , not exponentially. So spectacular ideas are rare but not vanishingly rare. A higher z_0 scales *all* ideas up proportionally; a higher θ makes the tail die faster, so ideas are *less dispersed* (more similar to each other). Keep this in mind: throughout EK, θ will always mean “inverse heterogeneity.”

Derivation: Density, mean, variance, coefficient of variation

Step 1: the density. Differentiate the CDF:

$$f(z) = F'(z) = \frac{d}{dz} \left[1 - z_0^\theta z^{-\theta} \right] = \theta z_0^\theta z^{-\theta-1}, \quad z \geq z_0.$$

Step 2: the mean. Requires $\theta > 1$ (otherwise the integral diverges):

$$\mathbb{E}[Z] = \int_{z_0}^{\infty} z f(z) dz = \theta z_0^\theta \int_{z_0}^{\infty} z^{-\theta} dz = \theta z_0^\theta \left[\frac{z^{1-\theta}}{1-\theta} \right]_{z_0}^{\infty} = \theta z_0^\theta \cdot \frac{z_0^{1-\theta}}{\theta-1} = \frac{\theta}{\theta-1} z_0.$$

(The bracket at ∞ vanishes because $1-\theta < 0$.)

Step 3: the second moment. Requires $\theta > 2$:

$$\mathbb{E}[Z^2] = \theta z_0^\theta \int_{z_0}^{\infty} z^{1-\theta} dz = \theta z_0^\theta \cdot \frac{z_0^{2-\theta}}{\theta-2} = \frac{\theta}{\theta-2} z_0^2.$$

Step 4: the variance.

$$\mathbb{V}[Z] = \mathbb{E}[Z^2] - \mathbb{E}[Z]^2 = z_0^2 \left[\frac{\theta}{\theta-2} - \frac{\theta^2}{(\theta-1)^2} \right] = z_0^2 \frac{\theta(\theta-1)^2 - \theta^2(\theta-2)}{(\theta-1)^2(\theta-2)}.$$

Expand the numerator: $\theta[(\theta - 1)^2 - \theta(\theta - 2)] = \theta[\theta^2 - 2\theta + 1 - \theta^2 + 2\theta] = \theta$. Hence

$$\mathbb{V}[Z] = \frac{\theta}{(\theta - 1)^2(\theta - 2)} z_0^2.$$

Step 5: coefficient of variation. Divide s.d. by mean:

$$CV = \frac{\sqrt{\mathbb{V}[Z]}}{\mathbb{E}[Z]} = \frac{z_0 \sqrt{\theta} / [(\theta - 1)\sqrt{\theta - 2}]}{\theta z_0 / (\theta - 1)} = \frac{\sqrt{\theta}}{\theta \sqrt{\theta - 2}} = [\theta(\theta - 2)]^{-1/2}.$$

Intuition: Reading the CV

The CV depends *only* on θ , not on z_0 . That is exactly the sense in which z_0 is pure scale and θ is pure dispersion: doubling z_0 doubles both the mean and the standard deviation, leaving relative spread unchanged. As $\theta \downarrow 2$ the CV explodes (huge heterogeneity); as $\theta \rightarrow \infty$ it goes to zero (all ideas essentially identical).

Check your understanding (Q1)

Without computing anything new: if $\theta = 3$ and we double z_0 , what happens to (a) $\mathbb{E}[Z]$, (b) $\mathbb{V}[Z]$, (c) the CV? (*Answers to all questions are in the Appendix.*)

1.1 Truncation from below preserves the Pareto shape

Derivation: Truncated Pareto is still Pareto

Condition on the idea being at least as good as some threshold $z_1 \geq z_0$. For $z \geq z_1$, by the definition of conditional probability (and using that $\{Z \geq z\} \subseteq \{Z \geq z_1\}$, so the intersection is just $\{Z \geq z\}$):

$$\Pr(Z \geq z \mid Z \geq z_1) = \frac{\Pr(Z \geq z)}{\Pr(Z \geq z_1)} = \frac{(z/z_0)^{-\theta}}{(z_1/z_0)^{-\theta}} = \left(\frac{z}{z_1}\right)^{-\theta}.$$

The z_0 's cancel: the conditional distribution is Pareto again, with the *same* shape θ and new scale z_1 .

Intuition: Scale invariance

Truncation from below just relabels the “floor”: among ideas better than z_1 , relative quality looks statistically identical to the original population. This self-similarity is the Pareto analogue of the exponential distribution’s memorylessness (indeed, $\log Z$ is exponential). It is why selection on productivity (only good ideas survive competition) will not destroy the distributional form later on.

Check your understanding (Q2)

A firm only observes ideas with $Z \geq z_1 = 2z_0$. Is the CV of the ideas it observes larger, smaller, or the same as the CV in the full population? Why?

2 The accumulation of ideas

Definition: Technology as a stock of ideas

Suppose N ideas have arrived, each with efficiency drawn i.i.d. from $\text{Pareto}(z_0, \theta)$. Define

$$N_z = \#\{\text{ideas with efficiency} > z\}.$$

“Technology” is the whole collection of arrived ideas; what production ultimately uses is the *best* of them.

Derivation: N_z is binomial, and its mean defines T

Each of the N ideas independently “succeeds” (exceeds z) with probability

$$p_z \equiv 1 - F(z) = \left(\frac{z}{z_0}\right)^{-\theta} = z_0^\theta z^{-\theta}.$$

A count of independent successes is binomial:

$$\Pr(N_z = n) = \binom{N}{n} p_z^n (1 - p_z)^{N-n}. \quad (2)$$

Its mean is the standard binomial mean Np_z :

$$\mathbb{E}[N_z] = Np_z = N z_0^\theta z^{-\theta} = T z^{-\theta}, \quad \boxed{T \equiv z_0^\theta N}.$$

Intuition: What is T ?

T is the **state of technology**: a single number combining *how many* ideas have arrived (N) and *how good* draws tend to be (z_0^θ). A country with double the T expects, at every quality cutoff z , twice as many ideas above that cutoff. T will become the country-level technology parameter T_i in EK, capturing R&D stocks, human capital, institutions, etc.

2.1 The Poisson limit

We now take a limit that keeps T fixed while letting $N \rightarrow \infty$ and $z_0^\theta \rightarrow 0$. Why? Letting $z_0 \rightarrow 0$ assigns positive probability to *all* $z > 0$ (arbitrarily bad ideas can arrive), and letting $N \rightarrow \infty$ means arbitrarily good ideas eventually arrive too. The product $T = z_0^\theta N$ stays meaningful.

Derivation: Binomial $\rightarrow$ Poisson

Fix z and let $\lambda \equiv p_z N = Tz^{-\theta}$. Substitute $p_z = \lambda/N$ into (2):

$$\Pr(N_z = n) = \frac{N!}{n!(N-n)!} \frac{\lambda^n}{N^n} \left(1 - \frac{\lambda}{N}\right)^{N-n} = \frac{\lambda^n}{n!} \underbrace{\left(1 - \frac{\lambda}{N}\right)^N}_{(a)} \underbrace{\frac{N!}{(N-n)! N^n}}_{(b)} \underbrace{\left(1 - \frac{\lambda}{N}\right)^{-n}}_{(c)}.$$

Take $N \rightarrow \infty$ holding λ and n fixed, term by term:

- (a) $\left(1 - \frac{\lambda}{N}\right)^N \rightarrow e^{-\lambda}$, using the classic limit $\lim_{N \rightarrow \infty} \left(1 + \frac{x}{N}\right)^N = e^x$ with $x = -\lambda$.
- (b) $\frac{N!}{(N-n)! N^n} = \frac{N(N-1) \cdots (N-n+1)}{N \cdot N \cdots N} = 1 \cdot \left(1 - \frac{1}{N}\right) \cdots \left(1 - \frac{n-1}{N}\right) \rightarrow 1$ (a fixed number n of factors, each $\rightarrow 1$).
- (c) $\left(1 - \frac{\lambda}{N}\right)^{-n} \rightarrow 1^{-n} = 1$.

Therefore

$$\lim_{N \rightarrow \infty} \Pr(N_z = n) = \frac{\lambda^n}{n!} e^{-\lambda} = \frac{(Tz^{-\theta})^n}{n!} e^{-Tz^{-\theta}}, \quad (3)$$

a **Poisson distribution** with rate $\lambda = Tz^{-\theta}$.

Intuition: Law of rare events

This is the “law of rare events”: many trials (N huge), each with tiny success probability (p_z small), with a stable expected count λ . Counting ideas above quality z is like counting radioactive decays or typos per page. The rate $\lambda = Tz^{-\theta}$ falls in z : the higher the quality bar, the fewer ideas clear it, and θ governs how fast that count thins out.

Check your understanding (Q3)

In the Poisson formula (3), what is the expected number of ideas with efficiency above z ? Compare it with the binomial mean $\mathbb{E}[N_z] = Tz^{-\theta}$ from before the limit. What do you conclude?

3 The most efficient idea: the Fréchet distribution

Derivation: CDF of the best idea

Rank ideas: $Z^{(1)} \geq Z^{(2)} \geq \dots$. The key logical step:

$$\{Z^{(1)} \leq z\} \iff \{\text{no idea exceeds } z\} \iff \{N_z = 0\}.$$

Plug $n = 0$ into the Poisson formula (3):

$$F^{(1)}(z) \equiv \Pr(Z^{(1)} \leq z) = \Pr(N_z = 0) = \frac{(Tz^{-\theta})^0}{0!} e^{-Tz^{-\theta}} = e^{-Tz^{-\theta}}.$$

This is the **Fréchet** (Type II extreme value) distribution.

Sanity check (a second route). Without the Poisson step, the max of N i.i.d. draws satisfies

$$\Pr(\max_i Z_i \leq z) = F(z)^N = (1 - p_z)^N = \left(1 - \frac{Tz^{-\theta}}{N}\right)^N \rightarrow e^{-Tz^{-\theta}} \text{ — same answer.}$$

Definition: Fréchet distribution

$$\Pr(Z \leq z) = F(z) = e^{-Tz^{-\theta}}, \quad z \geq 0, \quad (4)$$

with **scale** $T > 0$ and **shape** $\theta > 1$. Moments:

$$\mathbb{E}[Z] = \Gamma\left(1 - \frac{1}{\theta}\right) T^{1/\theta}, \quad \text{s.d.}(\log Z) = \frac{\pi}{\sqrt{6}} \frac{1}{\theta},$$

where $\Gamma(t) = \int_0^\infty x^{t-1} e^{-x} dx$ is the gamma function.

Derivation: Mean of a Fréchet random variable

The density is

$$f(z) = \frac{d}{dz} e^{-Tz^{-\theta}} = e^{-Tz^{-\theta}} \cdot \theta T z^{-\theta-1}.$$

So

$$\mathbb{E}[Z] = \int_0^\infty z \theta T z^{-\theta-1} e^{-Tz^{-\theta}} dz.$$

Substitute $x = Tz^{-\theta}$. Then $z = T^{1/\theta} x^{-1/\theta}$ and $dx = -\theta T z^{-\theta-1} dz$; as $z : 0 \rightarrow \infty$, $x : \infty \rightarrow 0$, and the sign flip reverses the limits:

$$\mathbb{E}[Z] = \int_0^\infty T^{1/\theta} x^{-1/\theta} e^{-x} dx = T^{1/\theta} \int_0^\infty x^{(1-1/\theta)-1} e^{-x} dx = T^{1/\theta} \Gamma\left(1 - \frac{1}{\theta}\right).$$

The integral converges only if $1 - 1/\theta > 0$, i.e. $\theta > 1$ — the Fréchet inherits the Pareto's fat tail, so low θ kills the mean.

For the log: one can show $\theta(\log Z - \frac{1}{\theta} \log T)$ has a Gumbel distribution, whose variance is $\pi^2/6$; hence $\text{s.d.}(\log Z) = \frac{\pi}{\sqrt{6}} \frac{1}{\theta}$. Again: **larger $\theta \Rightarrow$ less dispersion.**

Intuition: Why Fréchet is the star of the show

Extreme value theory says maxima of fat-tailed draws converge to Fréchet (just as sums converge to the normal). Because market competition selects the *best* idea, the distribution

of *used* technology is Fréchet, no matter how messy the underlying idea process, provided its tail is Pareto-like. Two parameters do all the work: T shifts the whole distribution right (better technology on average), θ compresses it (less scope for comparative advantage). In the pdf plot on slide 9: raising θ (with $T = 1$) makes the density taller and tighter; raising T (with $\theta = 2$) shifts mass to the right.

Check your understanding (Q4)

The best idea's CDF is $e^{-Tz^{-\theta}}$. What is the probability that the best idea is *better* than z when $Tz^{-\theta}$ is small? (Hint: $e^{-x} \approx 1 - x$.) Interpret.

4 Putting the technology block into a Ricardian model

Intuition: The powerful idea behind EK

Markets are perfectly competitive and all ideas for good ω produce perfectly substitutable varieties. Then only the *cheapest* producer of ω sells anything, at a price equal to the unit cost of the *most efficient* idea. So the technology actually observed in equilibrium is a *maximum* over ideas — and we just proved maxima are Fréchet. The whole trade model can therefore be written with two parameters per country (T_i , common θ), instead of a continuum of productivities.

4.1 Setup: DFS with probability theory

Two countries $i = H, F$ (Home, Foreign). Labor is the only factor, wage w_i . A continuum of goods $\omega \in [0, 1]$; varieties from H and F are perfect substitutes. Productivity (inverse unit labor requirement) $z_i(\omega)$ is the realization of $Z_i \sim \text{Fréchet}$:

$$F_i(z) = \Pr(Z_i \leq z) = e^{-T_i z^{-\theta}}, \quad z \in [0, \infty).$$

Iceberg trade costs $d_{ij} \geq 1$ ($d_{ii} = 1$): delivering one unit from i to j requires shipping d_{ij} units. Perfect competition $\Rightarrow$ country i can deliver good ω to country j at price $d_{ij}w_i/z_i(\omega)$.

Definition: Schedule of comparative advantage

As in Dornbusch–Fischer–Samuelson (DFS),

$$A(\omega) \equiv \frac{Z_H(\omega)}{Z_F(\omega)}.$$

The crucial difference: in DFS $A(\omega)$ is a *deterministic, downward-sloping function*; in EK it is a *random variable*, i.i.d. across goods. DFS asks “for which ω is $A(\omega) \geq d_{HF}w_H/w_F$?”; EK asks “with what *probability* is $A(\omega) \geq d_{HF}w_H/w_F$?” Every deterministic object in DFS becomes a distribution in EK.

4.2 Distribution of relative productivity

Derivation: CDF of $A(\omega) = Z_H/Z_F$, line by line

We want $\Pr(A \leq a) = \Pr(Z_H \leq aZ_F)$. Condition on the realization $Z_F = z_F$ and integrate over its density (law of total probability):

$$\Pr(A \leq a) = \int_0^\infty \Pr(Z_H \leq az_F) dF_F(z_F).$$

Ingredients. From the Fréchet CDFs:

$$\Pr(Z_H \leq az_F) = e^{-T_H(az_F)^{-\theta}}, \quad dF_F(z_F) = \theta T_F z_F^{-\theta-1} e^{-T_F z_F^{-\theta}} dz_F.$$

Substitute:

$$\Pr(A \leq a) = \int_0^\infty e^{-T_H a^{-\theta} z_F^{-\theta}} \theta T_F z_F^{-\theta-1} e^{-T_F z_F^{-\theta}} dz_F = \int_0^\infty \theta T_F z_F^{-\theta-1} e^{-(T_H a^{-\theta} + T_F) z_F^{-\theta}} dz_F.$$

(The two exponentials merged because they share the factor $z_F^{-\theta}$.)

Complete the density. Let $\Psi \equiv T_H a^{-\theta} + T_F$. The function $\theta \Psi z_F^{-\theta-1} e^{-\Psi z_F^{-\theta}}$ is exactly a Fréchet(Ψ, θ) density, so it integrates to 1. Multiply and divide by Ψ/T_F :

$$\Pr(A \leq a) = \frac{T_F}{\Psi} \underbrace{\int_0^\infty \theta \Psi z_F^{-\theta-1} e^{-\Psi z_F^{-\theta}} dz_F}_{=1} = \frac{T_F}{T_H a^{-\theta} + T_F}.$$

Intuition: A tug of war between technologies

$\Pr(A \leq a) = T_F/(T_H a^{-\theta} + T_F)$ is increasing in a (a CDF must be), increasing in T_F and decreasing in T_H : better Foreign technology makes low relative Home productivity more likely. The “complete the density” trick — recognizing the integrand as an unnormalized Fréchet density — is *the* workhorse move in this literature; you will use it three more times below.

4.3 Selection probabilities

Home serves the Foreign market in good ω iff its delivered price beats Foreign’s own cost: $d_{HF}w_H/Z_H \leq w_F/Z_F$, i.e. $A(\omega) \geq d_{HF}w_H/w_F$.

Derivation: Probability that F buys good ω from H

Using the CDF of A with $a = d_{HF}w_H/w_F$:

$$\pi_{HF} \equiv \Pr\left(A \geq \frac{d_{HF}w_H}{w_F}\right) = 1 - \frac{T_F}{T_H \left(\frac{d_{HF}w_H}{w_F}\right)^{-\theta} + T_F} = \frac{T_H \left(\frac{d_{HF}w_H}{w_F}\right)^{-\theta}}{T_H \left(\frac{d_{HF}w_H}{w_F}\right)^{-\theta} + T_F}.$$

Multiply numerator and denominator by $w_F^{-\theta}$:

$$\pi_{HF} = \frac{T_H(d_{HF}w_H)^{-\theta}}{T_H(d_{HF}w_H)^{-\theta} + T_Fw_F^{-\theta}}$$

Note π_{HF} does not depend on ω : it is the *same* for every good, so we can drop the argument.

Intuition: From probabilities to shares

π_{HF} is so far only a *probability* for one good. With a *continuum* of goods and independent draws, the Law of Large Numbers converts it into a deterministic *fraction*: exactly a share π_{HF} of goods (measure π_{HF} of the interval $[0, 1]$) is bought by F from H . Randomness at the good level, determinism in the aggregate. This is why EK can be estimated with aggregate trade data.

Check your understanding (Q5)

Suppose T_H doubles while everything else is constant. Does π_{HF} double? If not, in which direction does it move and why? What happens as $d_{HF} \rightarrow \infty$?

4.4 Distribution of prices

Derivation: Price offered by each source

The price of good ω in H if bought *from* H is $P_{HH} = w_H/Z_H$; a low price means a high Z . Its CDF:

$$F_{HH}(p) = \Pr(P_{HH} \leq p) = \Pr\left(\frac{w_H}{Z_H} \leq p\right) = \Pr\left(Z_H \geq \frac{w_H}{p}\right) = 1 - e^{-T_H(w_H/p)^{-\theta}} = 1 - e^{-T_Hw_H^{-\theta}p^\theta}.$$

Similarly if bought *from* F , $P_{FH} = d_{FH}w_F/Z_F$ and

$$F_{FH}(p) = \Pr\left(Z_F \geq \frac{d_{FH}w_F}{p}\right) = 1 - e^{-T_F(d_{FH}w_F)^{-\theta}p^\theta}.$$

These are *Weibull* distributions: the reciprocal of a Fréchet flips the fat tail from high productivity to low price.

Derivation: The price actually paid: distribution of the minimum

The buyer pays $P_H = \min\{P_{HH}, P_{FH}\}$. Use the complement trick — the min exceeds p iff *both* exceed p — plus independence:

$$\begin{aligned} F_H(p) &= \Pr(P_H \leq p) = 1 - \Pr(P_{HH} > p) \Pr(P_{FH} > p) \\ &= 1 - (1 - F_{HH}(p))(1 - F_{FH}(p)) \\ &= 1 - e^{-T_H w_H^{-\theta} p^\theta} e^{-T_F (d_{FH} w_F)^{-\theta} p^\theta} \\ &= 1 - e^{-\Phi_H p^\theta}, \quad \boxed{\Phi_H = T_H w_H^{-\theta} + T_F (d_{FH} w_F)^{-\theta}}. \end{aligned}$$

By symmetry, $F_F(p) = 1 - e^{-\Phi_F p^\theta}$ with $\Phi_F = T_H (d_{HF} w_H)^{-\theta} + T_F w_F^{-\theta}$.

Intuition: Φ : the market-access statistic

Everything a consumer in i cares about collapses into one number $\Phi_i = \sum_n T_n (d_{ni} w_n)^{-\theta}$: each source country contributes its technology T_n , *discounted* by its cost of serving you, $(d_{ni} w_n)^{-\theta}$. Trade benefits you by letting you tap other countries' T 's; distance and wages shrink the contribution; and θ governs how harshly costs are punished. Note also that the exponentials *multiplied* into a single one of the same form — minima of Weibulls are Weibull, the mirror image of maxima of Fréchet's being Fréchet. This closure property is what keeps the algebra tractable with any number of countries.

4.5 The price index

Preferences are CES over the continuum:

$$U_i = \left(\int_0^1 C_i(\omega)^{\frac{\sigma-1}{\sigma}} d\omega \right)^{\frac{\sigma}{\sigma-1}} \implies P_i = \left(\int_0^1 P_i(\omega)^{1-\sigma} d\omega \right)^{\frac{1}{1-\sigma}}.$$

By the LLN, integrating over goods equals integrating over the price distribution: $\int_0^1 P_i(\omega)^{1-\sigma} d\omega = \int_0^\infty p^{1-\sigma} dF_i(p)$.

Derivation: Computing the price index

The density from $F_i(p) = 1 - e^{-\Phi_i p^\theta}$ is $dF_i(p) = \theta \Phi_i p^{\theta-1} e^{-\Phi_i p^\theta} dp$. So

$$P_i^{1-\sigma} = \int_0^\infty p^{1-\sigma} \theta \Phi_i p^{\theta-1} e^{-\Phi_i p^\theta} dp.$$

Substitute $x = \Phi_i p^\theta$, so $p = (x/\Phi_i)^{1/\theta}$ and $dx = \theta \Phi_i p^{\theta-1} dp$:

$$P_i^{1-\sigma} = \int_0^\infty \left(\frac{x}{\Phi_i} \right)^{\frac{1-\sigma}{\theta}} e^{-x} dx = \Phi_i^{-\frac{1-\sigma}{\theta}} \int_0^\infty x^{(\frac{\theta+1-\sigma}{\theta}-1)} e^{-x} dx = \Phi_i^{-\frac{1-\sigma}{\theta}} \Gamma\left(\frac{\theta+1-\sigma}{\theta}\right).$$

Raise to the power $\frac{1}{1-\sigma}$:

$$\boxed{P_i = \gamma \Phi_i^{-1/\theta}, \quad \gamma = \left[\Gamma\left(\frac{\theta+1-\sigma}{\theta}\right) \right]^{\frac{1}{1-\sigma}}}$$

Convergence requires $\theta + 1 - \sigma > 0$, i.e. $\sigma < 1 + \theta$: substitutability across goods cannot be too strong relative to productivity heterogeneity, or the cheapest goods would dominate the index unboundedly.

Intuition: A striking simplification

The exact CES price index over a continuum of endogenously sourced goods is just a constant times $\Phi_i^{-1/\theta}$. Better access to world technology (higher Φ_i) means lower prices, with elasticity $1/\theta$. Remarkably, σ only affects the constant γ , not how prices respond to technology or trade costs — one reason the EK trade elasticity is θ , not $\sigma - 1$.

Check your understanding (Q6)

Where exactly does the requirement $\sigma < 1 + \theta$ come from in the derivation above? Which part of the price distribution causes trouble when it fails: very high prices or very low prices?

4.6 Equilibrium (two countries)

Combining the blocks: trade shares $\pi_{ij} = T_i(d_{ij}w_i)^{-\theta}/\Phi_j$, price indices $P_i = \gamma\Phi_i^{-1/\theta}$, and **labor market clearing**

$$w_i L_i = \sum_j \pi_{ij} w_j L_j :$$

country i 's labor income equals its sales to every market, where j 's total spending is its own labor income $w_j L_j$ (trade is balanced). Welfare per country is real income $U_i = w_i L_i / P_i$. Given $\{T_i, L_i, d_{ij}, \theta\}$, equilibrium is a wage vector solving these equations.

Part II

Eaton–Kortum (2002)

5 Background: why EK?

By the 1990s the Heckscher–Ohlin–Vanek (HOV) prediction — that countries export the services of their abundant factors — was known to fail badly in the data. Treffer (AER 1995) rejected HOV in favor of versions with home bias and international *technology* differences; Davis and Weinstein (AER 2001) rescued HOV only after adding technology differences, nontraded goods, and trade costs. The message: **technology differences and geography must be at the core of trade theory**. EK (2002) delivered exactly that, in a form (i) tractable with many countries, (ii) estimable with standard trade data, and (iii) usable for quantitative counterfactuals — launching the modern “quantitative trade” agenda.

6 Setup with N countries

Definition: EK environment

Countries $n = 1, \dots, N$; labor the only factor, fixed supply L_n ; continuum of goods $\omega \in [0, 1]$; perfect competition; constant returns to scale; iceberg costs $d_{in} \geq 1$, $d_{ii} = 1$, satisfying the **triangle inequality** $d_{ik}d_{kn} \geq d_{in}$ (no gains from re-routing through a third country — otherwise “ d_{in} ” would not be the true cost). Unit cost of delivering ω from i to j : $d_{ij}w_i/z_i(\omega)$. The price paid in j :

$$p_j(\omega) = \min \left\{ \frac{d_{ij}w_i}{z_i(\omega)}, i = 1, \dots, N \right\}.$$

CES preferences with elasticity σ . Technology: $Z_i(\omega)$ i.i.d. Fréchet, $F_i(z) = e^{-T_i z^{-\theta}}$, where T_i is country i 's **state of technology** (absolute advantage) and θ is (inverse) **heterogeneity in productivity** (scope for comparative advantage), common across countries.

7 Prices with N countries

Derivation: Price distributions, general N

Exactly as in the two-country case, for each source i ,

$$F_{ij}(p) \equiv \Pr(P_{ij} \leq p) = \Pr\left(Z_i \geq \frac{d_{ij}w_i}{p}\right) = 1 - e^{-T_i(d_{ij}w_i)^{-\theta}p^\theta},$$

and for the minimum over all N sources, using independence,

$$F_j(p) = 1 - \prod_{i=1}^N \Pr(P_{ij} > p) = 1 - \prod_{i=1}^N e^{-T_i(d_{ij}w_i)^{-\theta}p^\theta} = 1 - e^{-\Phi_j p^\theta}, \quad \Phi_j = \sum_{i=1}^N T_i(d_{ij}w_i)^{-\theta}.$$

The price index computation is verbatim the same as before:

$$P_j = \gamma \Phi_j^{-1/\theta}, \quad \gamma = \left[\Gamma\left(\frac{\theta + 1 - \sigma}{\theta}\right) \right]^{\frac{1}{1-\sigma}}. \quad (1)$$

8 Trade shares with N countries

The slides state the trade share; here is the full derivation, using the “complete the density” trick one more time.

Derivation: π_{ij} : probability that i is j 's cheapest supplier

$\pi_{ij} = \Pr(P_{ij} \leq \min_{k \neq i} P_{kj})$. Condition on $P_{ij} = p$ (density dF_{ij}) and require all rivals to be dearer:

$$\pi_{ij} = \int_0^\infty \prod_{k \neq i} \Pr(P_{kj} > p) dF_{ij}(p) = \int_0^\infty e^{-\Phi_j^{-i} p^\theta} \theta T_i(d_{ij}w_i)^{-\theta} p^{\theta-1} e^{-T_i(d_{ij}w_i)^{-\theta} p^\theta} dp,$$

where $\Phi_j^{-i} \equiv \sum_{k \neq i} T_k(d_{kj}w_k)^{-\theta} = \Phi_j - T_i(d_{ij}w_i)^{-\theta}$. Merge the exponentials ($\Phi_j^{-i} + T_i(d_{ij}w_i)^{-\theta} = \Phi_j$), then multiply and divide by Φ_j :

$$\pi_{ij} = \frac{T_i(d_{ij}w_i)^{-\theta}}{\Phi_j} \underbrace{\int_0^\infty \theta \Phi_j p^{\theta-1} e^{-\Phi_j p^\theta} dp}_{= 1 \text{ (a Weibull density)}}$$

$$\boxed{\pi_{ij} = \frac{T_i(d_{ij}w_i)^{-\theta}}{\sum_{\ell=1}^N T_\ell(d_{\ell j}w_\ell)^{-\theta}} = \frac{T_i(d_{ij}w_i)^{-\theta}}{\Phi_j}} \quad (2)$$

By the continuum + LLN argument, π_{ij} is also the *share of goods* — and, because the price distribution of what j actually buys from any source turns out to be the same $F_j(p)$ regardless of source, also the share of *expenditure* — that j devotes to goods from i .

Intuition: A gravity equation from first principles

Equation (2) is a *gravity equation*: i 's share in j 's spending rises with i 's technology T_i , falls with i 's wage and with bilateral distance d_{ij} , all relative to the toughness of competition in market j (Φ_j , an “inward multilateral resistance”). A subtle beautiful result lurking here: conditional on buying from *any* source, the price distribution is identical — goods that travel farther are not systematically more expensive, because only unusually productive draws survive long distances (selection exactly offsets the cost).

Definition: General equilibrium

Given $\{d_{ij}, T_i, L_i, \theta\}$, an equilibrium is a wage vector $\{w_i\}$ such that price indices satisfy (1), trade shares satisfy (2), and labor markets clear:

$$w_i L_i = \sum_j \pi_{ij} w_j L_j. \quad (3)$$

Welfare: $U_i = w_i L_i / P_i$.

Check your understanding (Q7)

Show, in one line each, that (i) $\sum_i \pi_{ij} = 1$ for every j , and (ii) equation (3) with π_{ij} summing to one implies world income equals world spending. Why is (i) economically necessary?

9 EK vs. Armington: what margin does trade adjust on?

Intuition: Two models, two margins, two elasticities

In **EK**, when a source becomes more remote (higher d_{ij}), it exports a *narrower range* of goods — the marginal goods in which its productivity edge was slim get undercut. Adjustment is on the **extensive margin**, and the elasticity of trade $X_{ij} = \pi_{ij} w_j L_j$ with respect

to d_{ij} is $-\theta$ (visible directly in equation (2)): a *technology-heterogeneity* parameter.

In **Armington**, each country is endowed with its own differentiated variety, so remoteness shrinks the *quantity* of the same variety — the **intensive margin** — with elasticity $1 - \sigma$: a *taste* parameter.

Same gravity form, completely different economics of the elasticity. In EK, more heterogeneity (low θ) means comparative advantage is strong, so trade responds *weakly* to costs; in Armington, more love of variety (low σ) does the analogous job.

Check your understanding (Q8)

In EK, why does *low* θ (more productivity heterogeneity) imply that trade flows respond *less* to trade costs? Answer with the selection logic, not the formula.

10 Estimation

10.1 The estimating equation

Derivation: From trade shares to gravity in logs

Let X_{ij} be j 's spending on goods from i , and X_j total spending, so $X_{ij} = \pi_{ij}X_j$. Divide j 's imports from i by j 's purchases from itself; Φ_j and X_j cancel:

$$\frac{X_{ij}}{X_{jj}} = \frac{\pi_{ij}}{\pi_{jj}} = \frac{T_i(d_{ij}w_i)^{-\theta}}{T_jw_j^{-\theta}} \quad (d_{jj} = 1).$$

Take logs:

$$\ln\left(\frac{X_{ij}}{X_{jj}}\right) = \underbrace{\ln T_i - \theta \ln w_i}_{\equiv S_i} - \underbrace{(\ln T_j - \theta \ln w_j)}_{= S_j} - \theta \ln d_{ij}.$$

S_i measures i 's **competitiveness**: technology net of wage costs — its potential to export gross of trade costs. Parameterize trade costs with observables (distance-bin dummies d_{ij}^k , shared border, shared language, RTA dummies, an importer effect m_j , and noise δ_{ij}):

$$\ln d_{ij} = m_j + d_{ij}^k + \text{border}_{ij} + \text{lang}_{ij} + \delta_{ij},$$

yielding the estimable two-way fixed-effects gravity equation

$$\ln\left(\frac{X_{ij}}{X_{jj}}\right) = S_i - S_j - \theta m_j - \theta d_{ij}^k - \theta \text{border}_{ij} - \theta \text{lang}_{ij} - \theta \delta_{ij}. \quad (4)$$

EK estimate (4) on 1990 manufacturing data for 19 OECD countries. Distance coefficients ($-\theta d_k$) fall monotonically from -3.10 (closest bin) to -6.56 (farthest); shared border and language raise trade.

Intuition: The identification problem

Equation (4) identifies only *products*: θd_k , θb , θl , and the composites S_i . It cannot separate θ from the trade-cost parameters — a world with mild cost sensitivity but huge barriers looks identical to one with brutal sensitivity and small barriers. Yet θ alone is the key welfare parameter. So EK need *outside* variation.

10.2 Strategy 1: unpack S_i with technology proxies

Since $S_i = \ln T_i - \theta \ln w_i$, regress estimated competitiveness on determinants of T_i and on wages:

$$\hat{S}_i = \alpha_0 + \alpha_R \ln R_i + \alpha_H \ln H_i - \theta \ln w_i + \epsilon_i,$$

with R_i = R&D stock, H_i = human capital proxy. The coefficient on $\ln w_i$ recovers $-\theta$.

Check your understanding (Q9)

Why is OLS on this regression problematic? (This is the slide's own "[Why?]"). And why might total workforce and population density be usable instruments for the wage? (The slide's "[How?]").

Results: OLS gives $\hat{\theta} = 2.84$; instrumenting the wage with workforce and population density gives $\hat{\theta} = 3.60$.

10.3 Strategy 2: retail price data

EK collected retail prices of 50 manufactured products in each of the 19 countries. Let $r_{ij}(\omega) = \ln p_j(\omega) - \ln p_i(\omega)$.

Derivation: Bounding trade costs with no-arbitrage

No-arbitrage: j can always import ω from i at cost $p_i(\omega)d_{ij}$, so $p_j(\omega) \leq p_i(\omega)d_{ij}$, i.e.

$$r_{ij}(\omega) \leq \ln d_{ij} \quad \text{for every } \omega.$$

So the *maximum* of $r_{ij}(\omega)$ over goods estimates $\ln d_{ij}$ (goods j actually imports from i sit *at* the bound). To be robust to measurement error, EK use the *second*-highest value, $\max_2$. Their proxy for $\ln(p_i d_{ij}/p_j)$ — trade-cost-adjusted relative price levels — is

$$D_{ij} = \max_2 \{r_{ij}(\omega)\} - \frac{1}{50} \sum_{\omega=1}^{50} r_{ij}(\omega),$$

the gap between the (near-)extreme relative price and the average relative price. In the model, $\ln(X_{ij}/X_j) - \ln(X_{jj}/X_j)$ should equal $-\theta \times$ this quantity. Plotting normalized import shares against D_{ij} and fitting the slope yields

$$\hat{\theta} = 8.28,$$

EK's preferred estimate (with 3.60 and 12.86 as low/high alternatives).

Intuition: Why price dispersion identifies θ

θ says how much trade flows respond to cost differences. Trade data alone give the response of flows; price data give the cost differences themselves. Combining them pins down the ratio. Bigger measured price gaps with the same trade response $\Rightarrow$ smaller θ ; tiny price gaps generating big trade responses $\Rightarrow$ huge θ .

11 Calibration and counterfactuals

For quantitative realism EK add (i) a *nontraded* sector (with labor either mobile or immobile between it and manufacturing), and (ii) *intermediate inputs*: manufacturing uses manufactures as inputs, so input costs are not just wages but a cost bundle involving the price index itself — amplifying trade’s effects.

Counterfactual 1: autarky. Raising $d_{ni} \rightarrow \infty$ for all $n \neq i$: welfare losses range from -0.2% (Japan, large and remote already) to -10.3% (Belgium, small and open). Small open economies lose most; manufacturing prices rise everywhere.

Counterfactual 2: zero gravity / doubled trade. Removing all geographic barriers raises welfare by 16–24% (large gains everywhere); doubling trade volumes yields 1–3%. Labor reallocates across sectors, with technology leaders (Germany, Japan, US) *shrinking* manufacturing employment under zero gravity as they import more varieties cheaply.

Check your understanding (Q10)

Using $P_j = \gamma \Phi_j^{-1/\theta}$ and $\pi_{jj} = T_j w_j^{-\theta} / \Phi_j$, show that real income can be written

$$\frac{w_j}{P_j} = \gamma^{-1} \left(\frac{T_j}{\pi_{jj}} \right)^{1/\theta} w_j^0 \quad \text{— more precisely, } \frac{w_j}{P_j} = \gamma^{-1} T_j^{1/\theta} \pi_{jj}^{-1/\theta}.$$

Conclude: how do the gains from trade (relative to autarky, where $\pi_{jj} = 1$) depend only on π_{jj} and θ ? Why does a *larger* θ imply *smaller* gains from trade for a given home share?

A Answers to the checkpoint questions

Answer Q1 (Pareto moments)

(a) $\mathbb{E}[Z] = \frac{\theta}{\theta-1} z_0$ doubles. (b) $\mathbb{V}[Z] \propto z_0^2$ quadruples. (c) $CV = [\theta(\theta-2)]^{-1/2}$ is unchanged — it depends only on θ . Scale in, scale out.

Answer Q2 (truncation)

The same. Truncation from below yields Pareto with the *same* θ (only the scale changes from z_0 to z_1), and the CV depends only on θ . Relative dispersion among surviving ideas is identical to the population — the self-similarity property.

Answer Q3 (Poisson mean)

A $\text{Poisson}(\lambda)$ variable has mean $\lambda = Tz^{-\theta}$ — exactly the binomial mean Np_z before the limit. The limit was engineered (holding $T = z_0^\theta N$ fixed) to preserve the expected number of ideas above every cutoff while smoothing the distribution into Poisson form.

Answer Q4 (Fréchet tail)

$\Pr(Z^{(1)} > z) = 1 - e^{-Tz^{-\theta}} \approx Tz^{-\theta}$ for small $Tz^{-\theta}$. Far in the upper tail, the best idea inherits a Pareto-like tail with the same θ , scaled by T : the probability the frontier exceeds z is (approximately) just the expected number of ideas above z , because at most one idea is likely to be up there.

Answer Q5 (comparative statics of π_{HF})

No. Write $\pi_{HF} = \frac{x}{x+y}$ with $x = T_H(d_{HF}w_H)^{-\theta}$, $y = T_Fw_F^{-\theta}$. Doubling T_H doubles x but also raises the denominator, so π_{HF} rises *less* than proportionally (it is concave in T_H and bounded by 1) — Home increasingly competes against itself for marginal goods. As $d_{HF} \rightarrow \infty$, $x \rightarrow 0$ so $\pi_{HF} \rightarrow 0$: prohibitive costs kill all exports (extensive margin shuts down good by good).

Answer Q6 (convergence of the price index)

It comes from the gamma-function integral $\int_0^\infty x^{\frac{\theta+1-\sigma}{\theta}-1} e^{-x} dx$, which converges at $x = 0$ only if the exponent $\frac{\theta+1-\sigma}{\theta} > 0$. Since $x = \Phi p^\theta$, $x \rightarrow 0$ corresponds to $p \rightarrow 0$: the problem is *very low* prices. With σ large, consumers pile onto the cheapest goods; $p^{1-\sigma}$ then blows up faster near $p = 0$ than the density vanishes, and the index diverges.

Answer Q7 (shares sum to one; adding up)

(i) $\sum_i \pi_{ij} = \sum_i T_i (d_{ij} w_i)^{-\theta} / \Phi_j = \Phi_j / \Phi_j = 1$: every good bought by j is bought from *somewhere* — the events “ i is the cheapest source” partition the probability space (ties have probability zero under continuous distributions). (ii) Summing (3) over i : $\sum_i w_i L_i = \sum_j (\sum_i \pi_{ij}) w_j L_j = \sum_j w_j L_j$ — world income = world spending, so Walras’s law holds and one market-clearing equation is redundant (a numeraire can be chosen).

Answer Q8 (low θ , low trade elasticity)

With high heterogeneity (low θ), the goods a country exports are ones where its productivity draw is typically *far* better than rivals’. A small increase in trade cost then dislodges very few goods — the marginal good is rare because efficiency gaps are wide. With low heterogeneity (high θ), exporters win by razor-thin margins, so tiny cost changes reshuffle many goods: trade is highly elastic.

Answer Q9 (OLS and instruments)

Why OLS fails: ϵ_i contains the part of technology T_i not captured by R_i, H_i . But wages are an equilibrium outcome: countries with better unmeasured technology have higher labor demand, hence higher w_i . So $\ln w_i$ is positively correlated with ϵ_i , biasing $\hat{\theta}$ (attenuating the wage coefficient). *Why the instruments:* total workforce and population density shift the labor *supply* side — they move wages (relevance: bigger labor supply, lower wages) but plausibly do not enter ϵ_i directly, i.e. they affect competitiveness only *through* wages, not through the state of technology (exclusion; debatable, but that is the argument).

Answer Q10 (gains from trade)

From $\pi_{jj} = T_j w_j^{-\theta} / \Phi_j$ we get $\Phi_j = T_j w_j^{-\theta} / \pi_{jj}$. Substitute into $P_j = \gamma \Phi_j^{-1/\theta}$:

$$P_j = \gamma (T_j w_j^{-\theta} / \pi_{jj})^{-1/\theta} = \gamma T_j^{-1/\theta} w_j \pi_{jj}^{1/\theta} \implies \frac{w_j}{P_j} = \gamma^{-1} T_j^{1/\theta} \pi_{jj}^{-1/\theta}.$$

Relative to autarky ($\pi_{jj} = 1$), the gain is $\pi_{jj}^{-1/\theta}$: it depends *only* on the observed home expenditure share and θ (the ACR formula). Larger θ shrinks gains for a given π_{jj} : with little heterogeneity, foreign goods are close substitutes for what you could make yourself, so access to them is worth less. This is why estimating θ correctly matters so much — moving from $\theta = 3.60$ to 12.86 cuts measured gains dramatically.